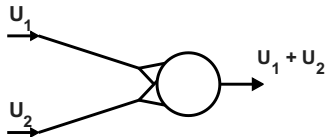
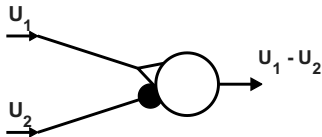
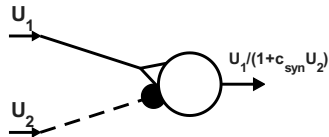


(a) addition

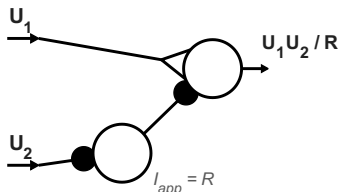
two excitatory signal-transmission synapses

(b) subtraction

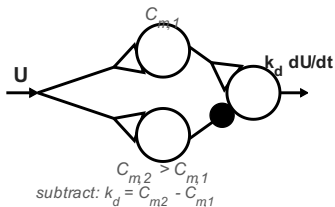
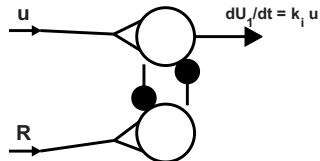
inhibitory synapse weight = $-k_{\text{syn},1} \Delta E_{s,1} / \Delta E_{s,2}$

(c) division

shunting: $\Delta E_{s,2} = 0$; dashed = modulatory

(d) multiplication

disinhibitory cascade: modulatory synapses in series

(e) differentiation**(f) integration**

mutual disinhibition = line attractor (no leak)